CPE112RC — Module1 — Linear Data Structures

Lecture2: Array/2D-Array/Matrix

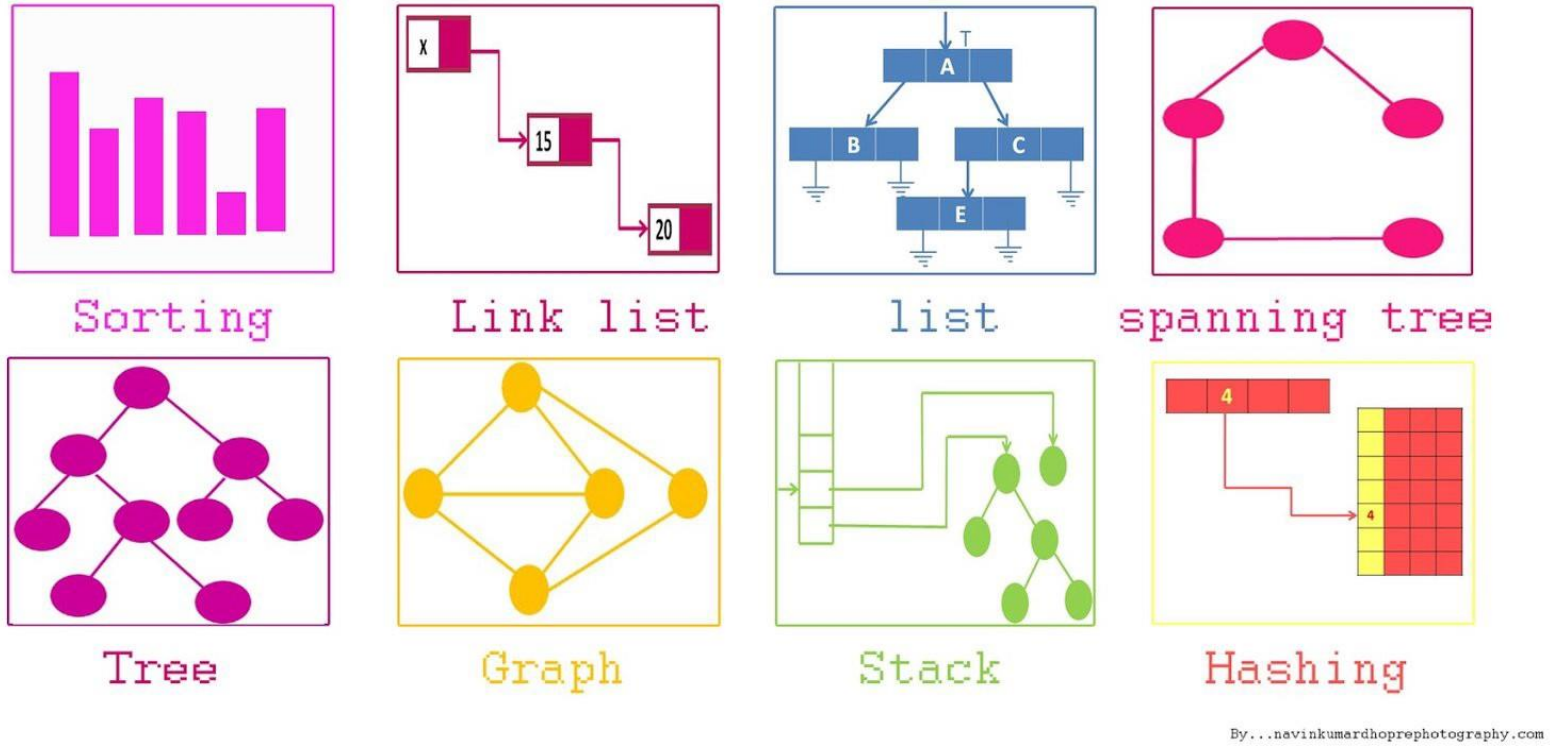
Dr. Prapong Prechaprapranwong

Computer Engineering

King Mongkut's University of Technology Thonburi

Abstract Data Type (ADT) = A definition for a data type

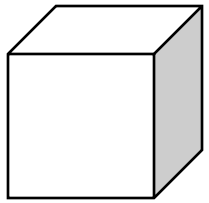
- a set of values
- a set of Operations allowed on data type



Content

- Array ADT
  - Array v.s. Python List
  - Behind dynamic array
- 2D-Array ADT
- Matrix ADT
  - Matrix Operations

- 0-dimension

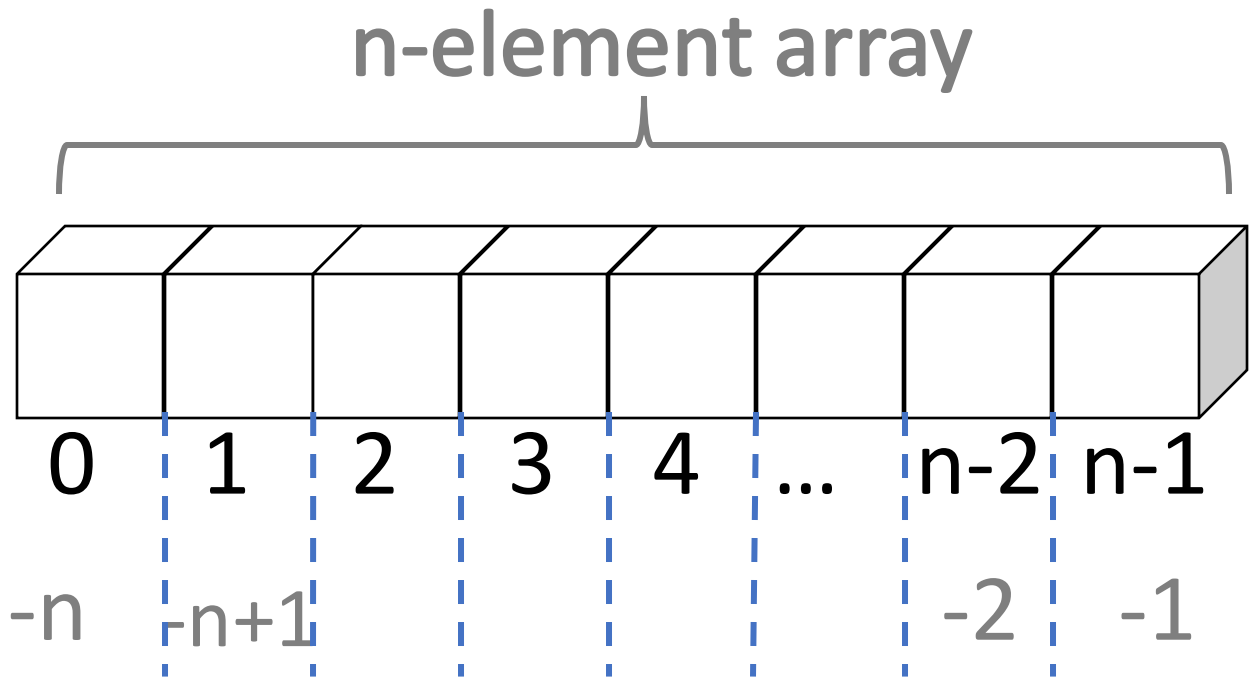


0-D arrays = Scalars
= value of the elements in an array.

Example:

```
a = numpy.array(100)
```

- 1-dimension



1-D arrays

= Uni-dimensional

→ positive index

← negative index

e.g.; time-series data

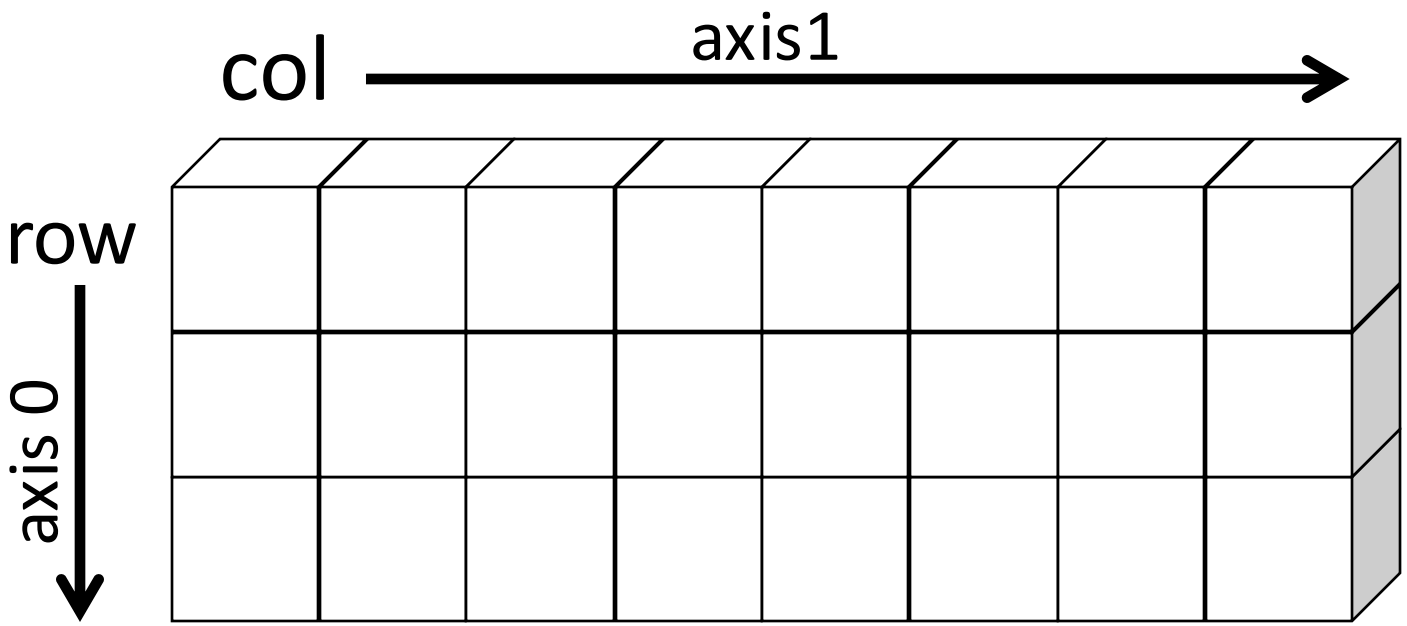
Example:

```
a = numpy.array([1, 2, 3, 4, 5])
```

contain 1-D arrays as elements

= 2-D arrays, 2nd order tensors or Matrix

- 2-dimension



4k = 3840 * 2160

2k = 2560 * 1440 = 1440p

2k = 1920 * 1080 = 1080p

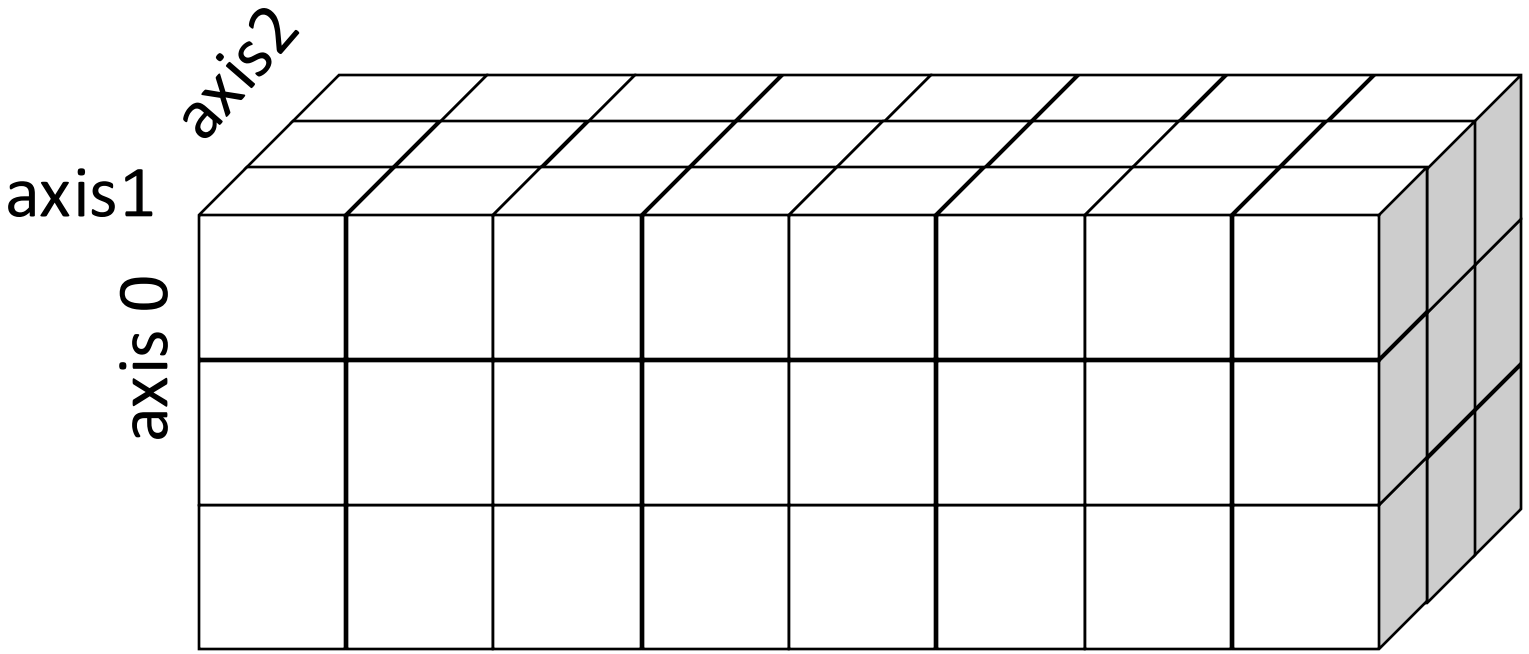
e.g.; image

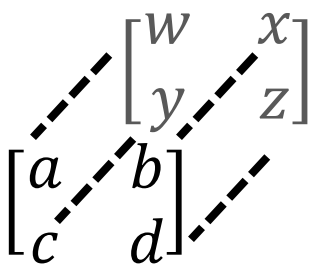
Example:

```
a = numpy.array([[1, 2], [10, 20]])
a = numpy.matrix ([[1, 2], [10, 20]])
```

- 3-dimension

contain 2-D arrays (matrices)
as elements = 3-D arrays
also called 3rd order tensor





e.g.; VDO files

Example:

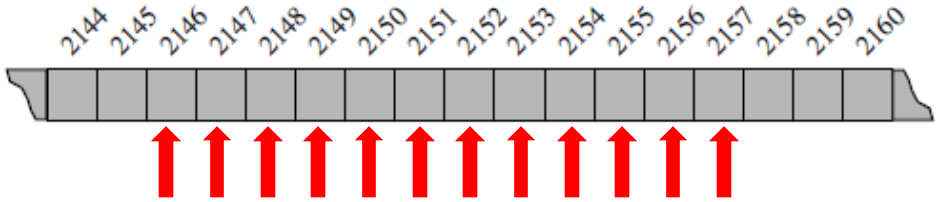
```
a = numpy.array([[[1,2],[10,20]],
                 [[3,4],[40,50]]])
```

Array ADT

A one-dimensional array is a collection of contiguous elements in which individual elements are identified by a unique integer subscript starting with zero. Once an array is created, its size cannot be changed.

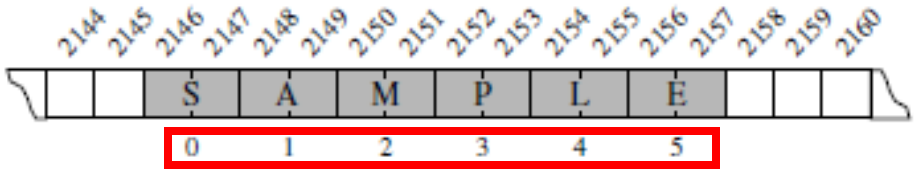
Array(size):	Creates a one-dimensional array consisting of size elements with each element initially set to None. size must be greater than zero
length ():	Returns the length or number of elements in the array.
__getitem__ (index):	Returns the value stored in the array at element position index. The Index argument must be within the valid range. Accessed using the subscript operator
__setitem__ (index, value):	Modifies the contents of the array element at position index to contain value. The index must be within the valid range. Accessed using the subscript operator.
clear(value):	Clears the array by setting every element to value
iterator ():	Creates and returns an iterator that can be used to traverse the elements of the array

Data stored in Memory address



1char: 2 bytes

a n a r r a y o f s i x c h a r a c t e r s , requires 12 bytes of memory. We will refer to each location within an array as a cell, and will use an **integer index** to describe its location within the array



integer index

sequential data type

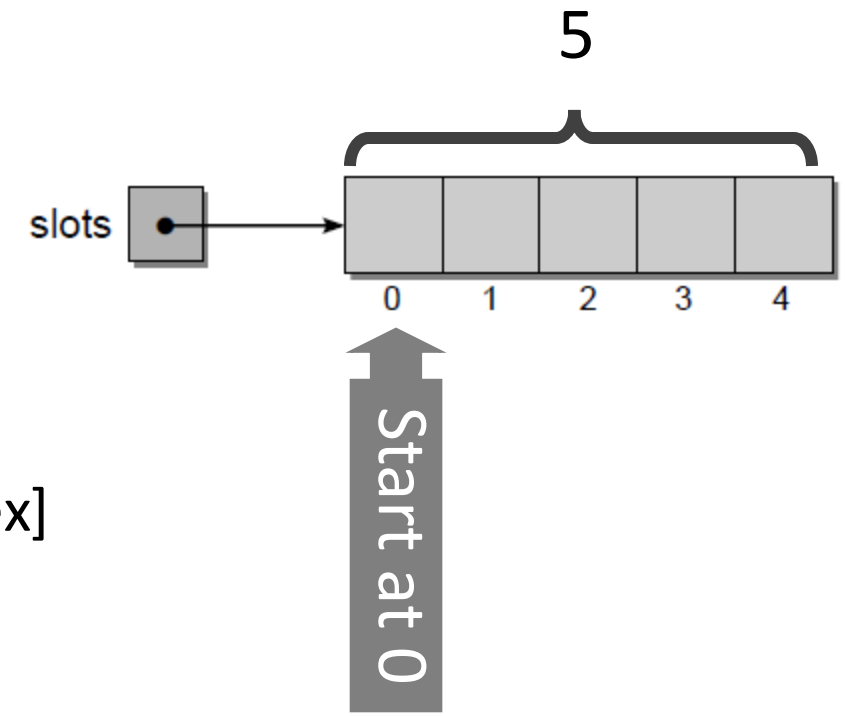
S	A	M	P	L	E
0	1	2	3	4	5

higher-level abstraction

array(size) - Create array with size elements

```
import ctypes

ArrayType = ctypes.py_object * 5
slots = ArrayType()
```



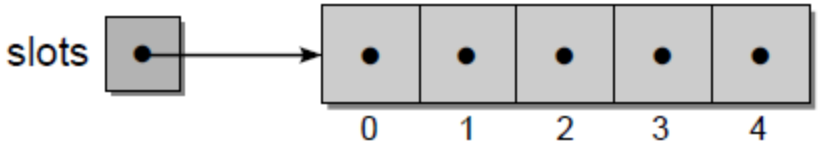
__getitem__(index) - recall the value at slots[index]

```
print( slots[0] )
```

→ Error

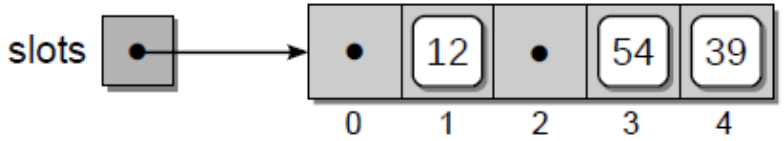
clear(None) - assign all value to **N o n e**

```
for i in range( 5 ) :
  slots[i] = None
```



__setitem__(index,value) - assign other values

```
slots[1] = 12
slots[3] = 54
slots[4] = 39
```



This module provides access to the diverse set of data types available in the C language and the complete functionality provided by a wide range of C libraries.

```
# Implements the Array ADT using array capabilities of the ctypes module.
import ctypes

class Array :
   # Creates an array with size elements.
  def __init__( self, size ):
    assert size > 0, "Array size must be > 0"
    self._size = size
     # Create the array structure using the ctypes module.
    PyArrayType = ctypes.py_object * size
    self._elements = PyArrayType()
     # Initialize each element.
    self.clear( None )

   # Returns the size of the array.
  def __len__( self ):
    return self._size
```

check status, “Error message”

Constructor

The __len__method, which returns the number of elements in the array, simply returns the value of size that was saved in the constructor.

the __getitem__ operator method takes the array index as an argument and returns the value of the corresponding element.

```
  # Gets the contents of the index element.
def __getitem__( self, index ):
  assert index >= 0 and index < len(self), "Array subscript out of range"
  return self._elements[ index ]

  # Puts the value in the array element at index position.
def __setitem__( self, index, value ):
  assert index >= 0 and index < len(self), "Array subscript out of range"
  self._elements[ index ] = value

  # Clears the array by setting each element to the given value.
def clear( self, value ):
  for i in range( len(self) ) :
    self._elements[i] = value
```

check status, “Error message”

the __setitem__operator method is used to set or change the contents of a specific element of the array

Python List →Dynamic Array

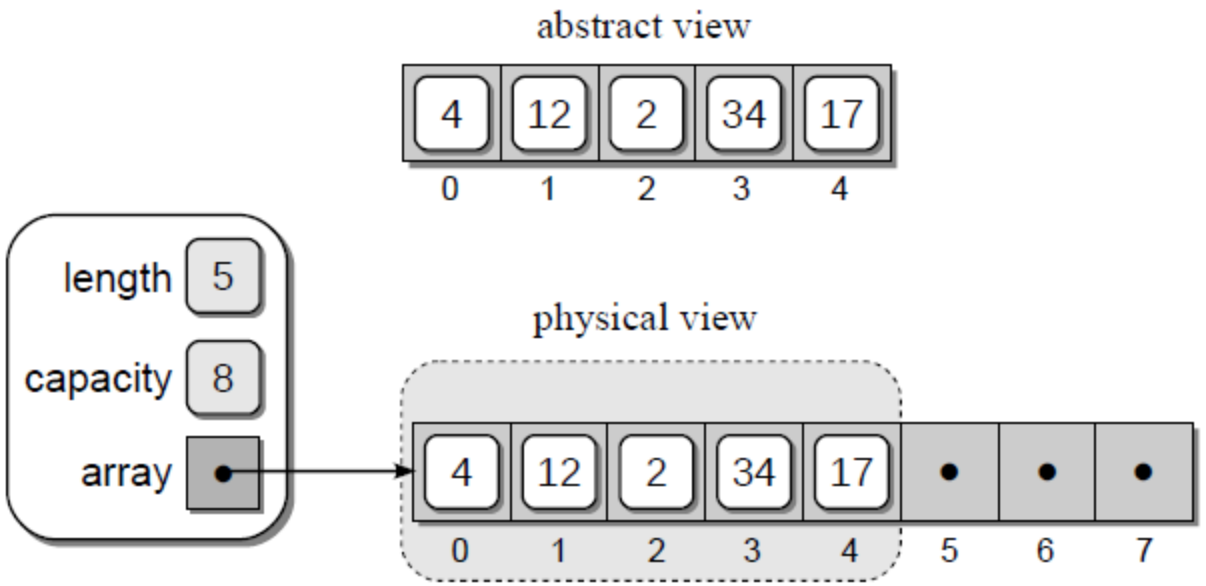
```
import sys                                    # provides getsizeof function
data = [ ]
for k in range(n):                            # NOTE: must fix choice of n
  a = len(data)                               # number of elements
  b = sys.getsizeof(data)                     # actual size in bytes
  print('Length: {0:3d}; Size in bytes: {1:4d}'.format(a, b))
  data.append(None)                           # increase length by one
```

Array vs Python List

- an array has a limited number of operations
- The array is suited when the <u>maximum no. of elements are known</u>
- The list is the better choice when the size may change after it created
- Python list will use four to five time as much memory

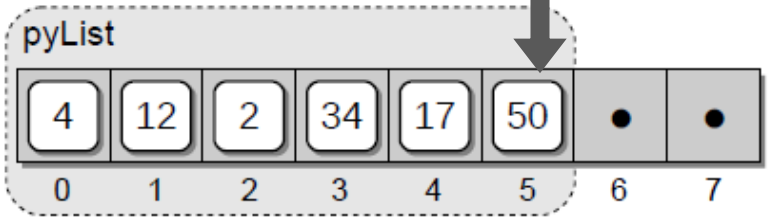
Behind the scene - Python List

```
pyList = [ 4, 12, 2, 34, 17 ]
```

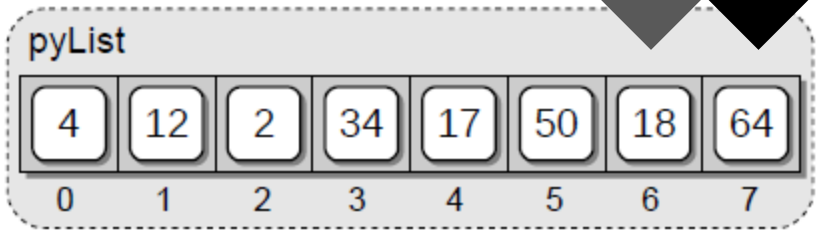


Append the list

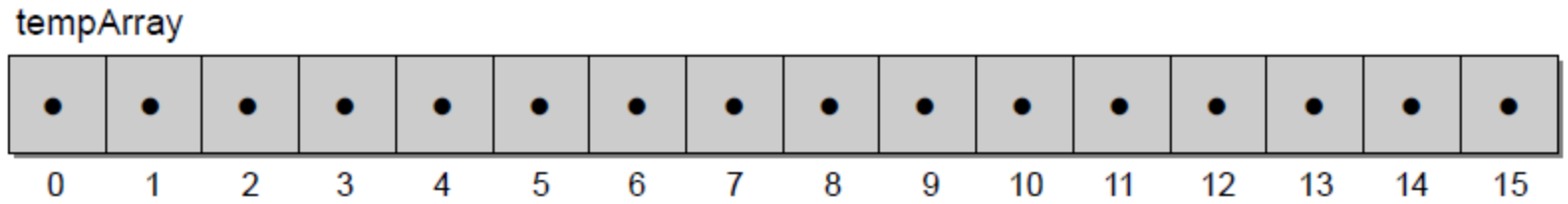
```
pyList.append( 50 )
```



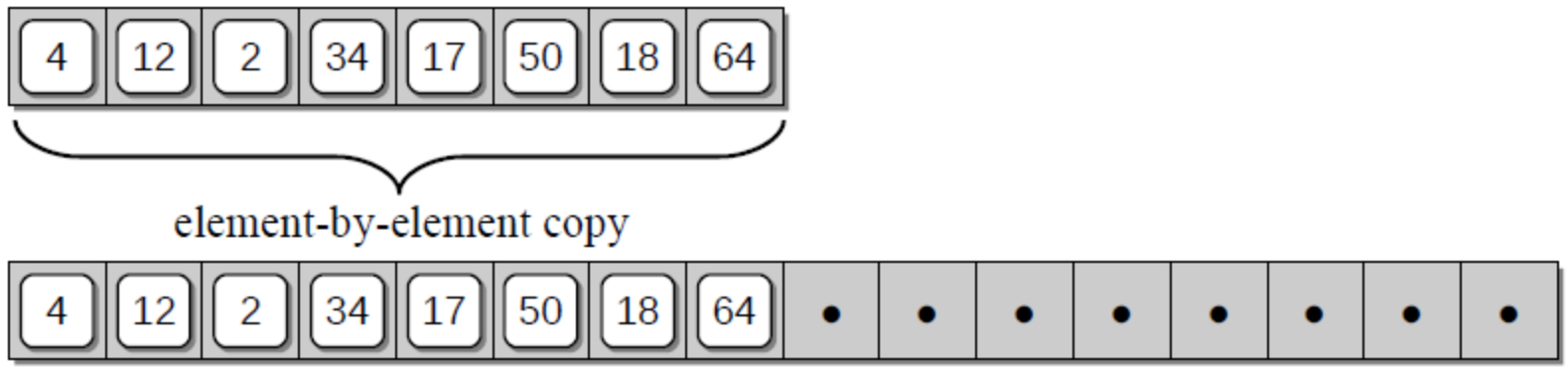

```
pyList.append( 18 )
pyList.append( 64 )
pyList.append( 6 )
```



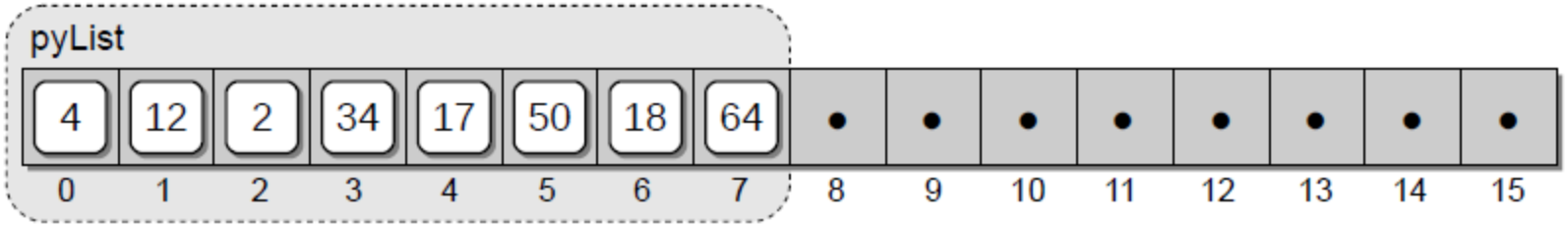
(1) A new array, double the size of the original, is created.



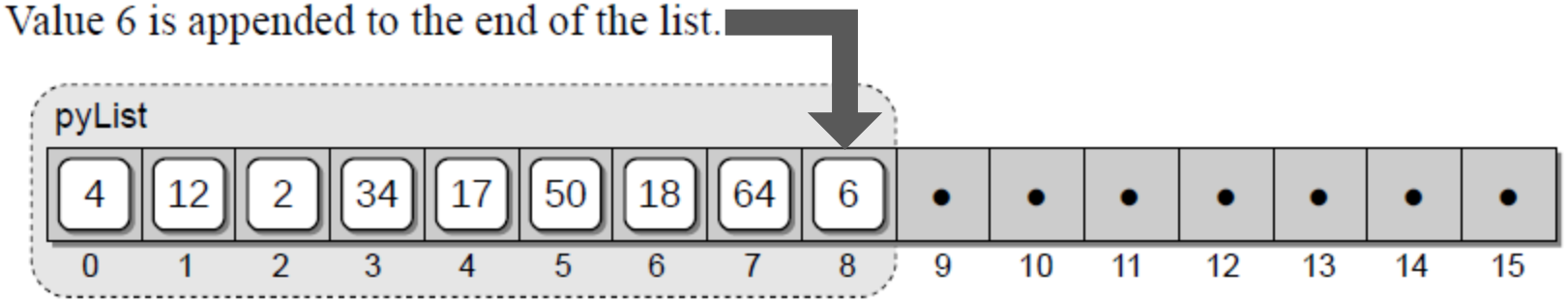
(2) The values from the original array are copied to the new larger array.



(3) The new array replaces the original in the list.

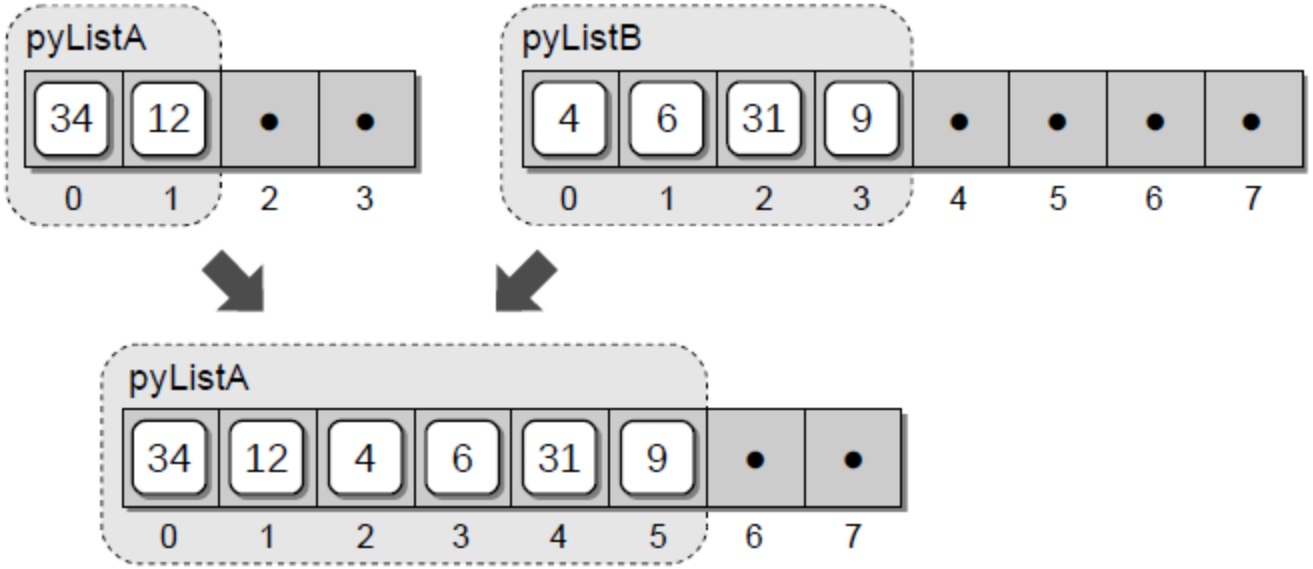


(4) Value 6 is appended to the end of the list.



Extend the list

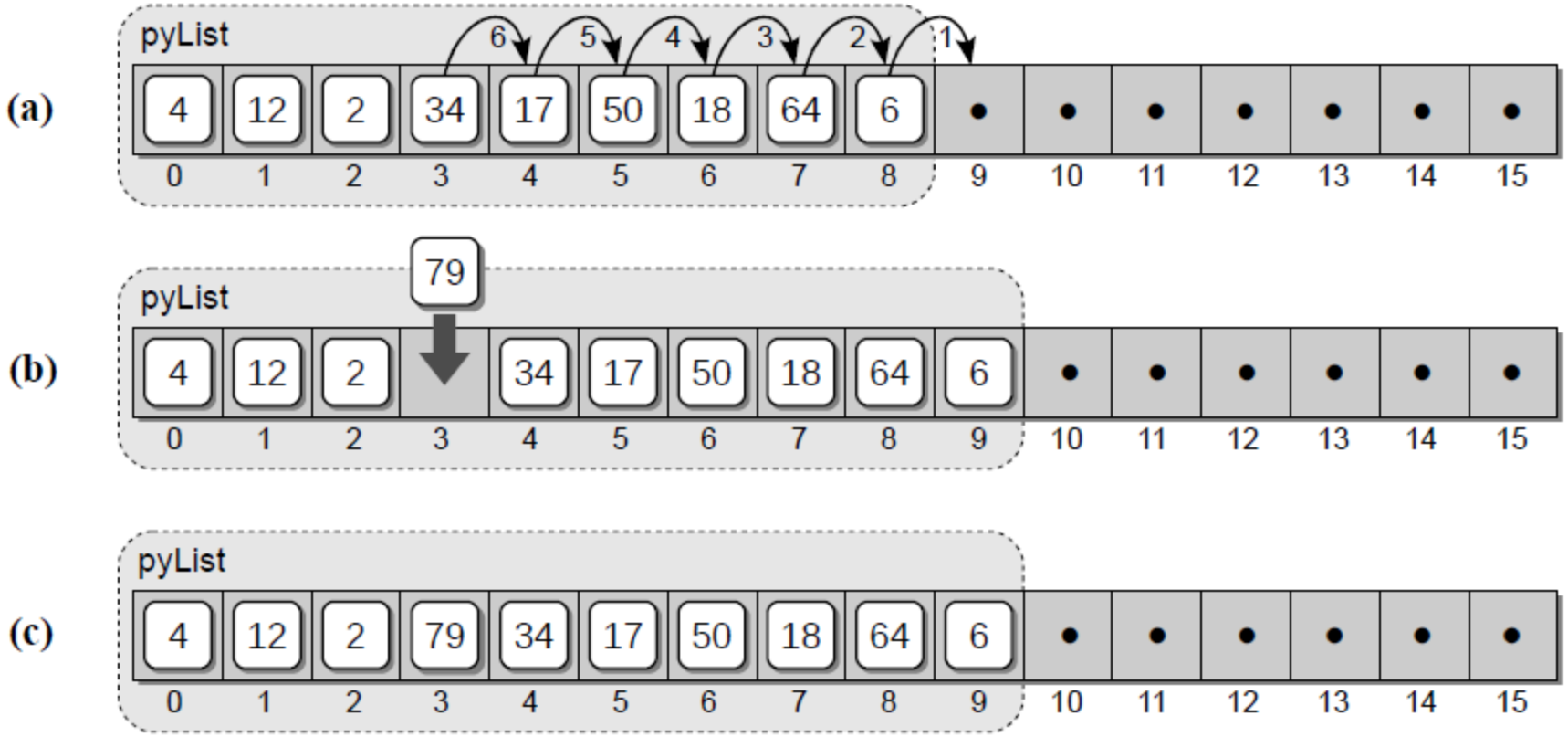
```
pyListA = [ 34, 12 ]
pyListB = [ 4, 6, 31, 9 ]
pyListA.extend( pyListB )
```



Insert an element to the list

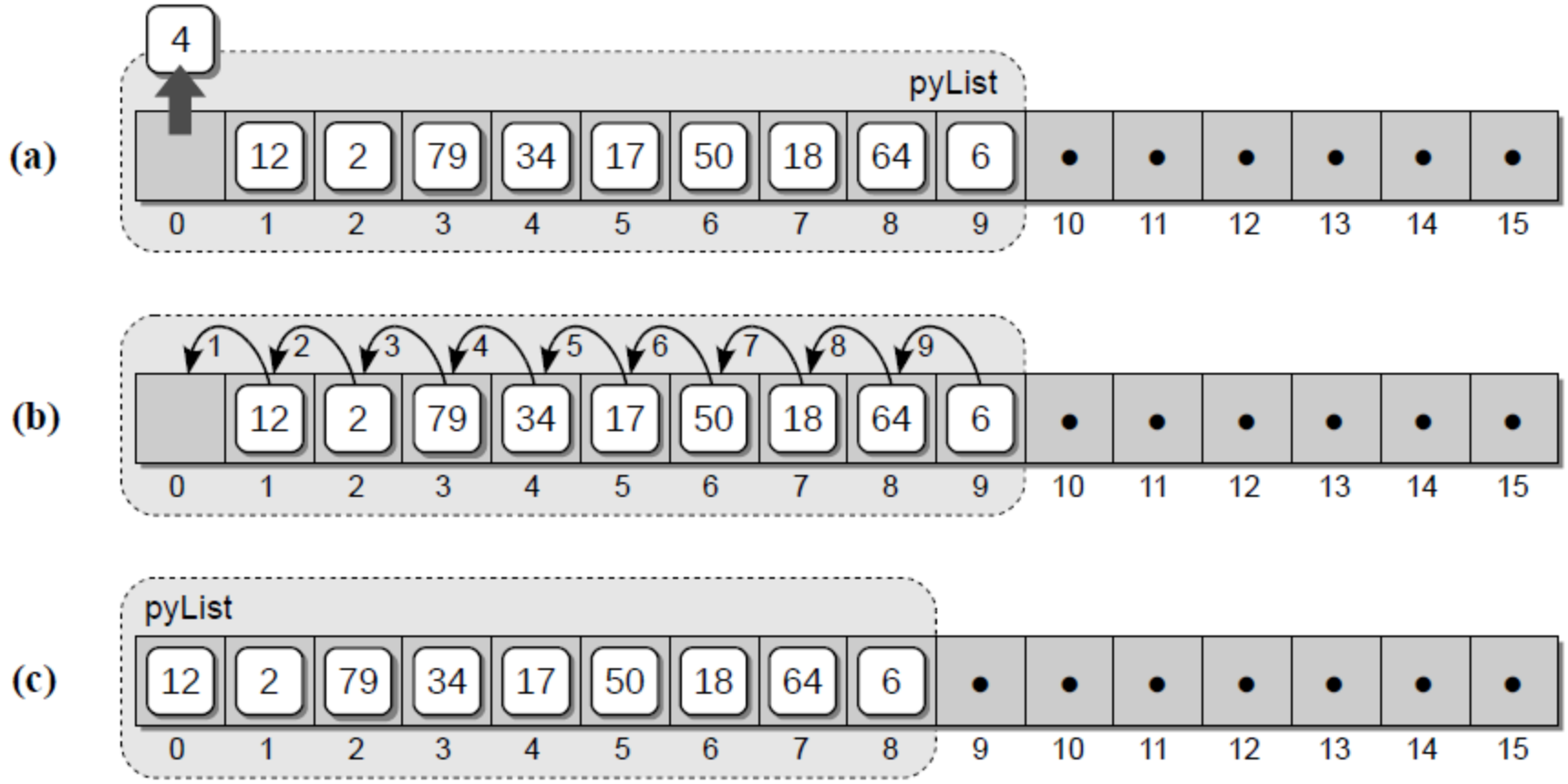
```
pyList.insert( 3, 79 )
```

at position:3 with value:79



pop an element from the list

```
pyList.pop( 0 )    # remove the first item
pyList.pop()       # remove the last item
```



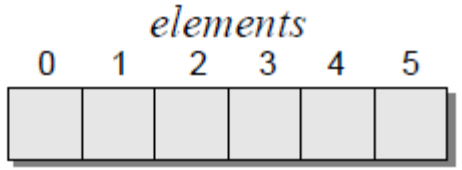
Array2D ADT

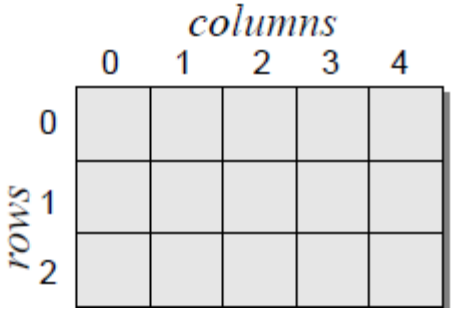
A two-dimensional array consists of a collection of elements organized into rows and columns. Individual elements are referenced by specifying the specific row and column indices (r,c), both of which start at 0.

Array2D(nrows, ncols):	Creates a two-dimensional array organized into rows and columns. The **nrows** and **ncols** indicate the size of the table. The individual elements of the table are initialized to None
numRows():	Returns the number of rows in the 2-D array
numCols():	Returns the number of columns in the 2-D array
clear(value):	Clears the array by setting each element to the given value
__getitem__(i1, i2):	Returns the value stored in the 2-D array element at the position indicated by the 2-tuple (i1; i2), both of which must be within the valid range. Accessed using the subscript operator: y = x[1,2]
__setitem__(i1, i2, value):	Modifies the contents of the 2-D array element indicated by the 2-tuple (i1; i2) with the new value. Both indices must be within the valid range. Accessed using the subscript operator: x[0,3] = y

The 2D-Array

A2D = Array2D(3,5)

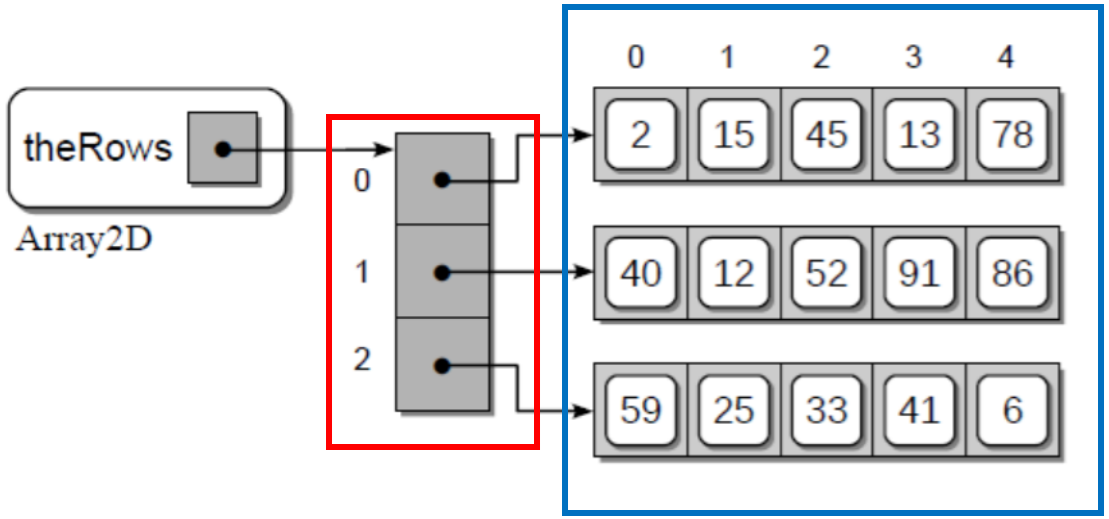





```python
class Array2D:
    def __init__(self,numRows,numCols):
        self._theRows = Array(numRows)
        for i in range(numRows):
            self._theRows[i] = Array(numCols)
```

2-dimensional array creates from a number of 1-dimensional arrays (theRows)

	0	1	2	3	4
0	2	15	45	13	78
1	40	12	52	91	86
2	59	25	33	41	6



```python
def __getitem__(self,indexTuple): # Receive 2 parameters as 1 Tuple
    assert len(indexTuple) == 2, "Invalid number of array subscripts."
    row = indexTuple[0]
    col = indexTuple[1]
    assert (row >= 0) and (row < self.numRows()) \
        and (col >= 0) and (col < self.numCols()),\
            "Array subscript out of range."
    return self._theRows[row][col]

def __setitem__( self, indexTuple, value ):
    assert len(indexTuple) == 2, "Invalid number of array subscripts."
    row = indexTuple[0]
    col = indexTuple[1]
    assert row >= 0 and row < self.numRows()\
        and col >= 0 and col < self.numCols(),\
            "Array subscript out of range."
    self._theRows[row][col] = value
```

Matrix ADT

A matrix is a collection of scalar values arranged in rows and columns as a rectangular grid of a fixed size. The elements of the matrix can be accessed by specifying a given row and column index with indices starting at 0.

Matrix(rows, ncols):	Creates a new matrix containing **nrows** and **ncols** with each element initialized to 0
numRows():	Returns the number of rows in the matrix
numCols():	Returns the number of columns in the matrix
__getitem__(row, col):	Returns the value stored in the given matrix element. Both row and col must be within the valid range
__setitem__ (row, col, scalar):	Sets the matrix element at the given row and col to scalar. The element indices must be within the valid range

Additional operations

scaleBy(scalar):	Multiplies each element of the matrix by the given **scalar** value. The matrix is modified by this operation
transpose():	Returns a new matrix that is the transpose of this matrix
add (rhsMatrix):	Creates and returns a new matrix that is the result of adding this matrix to the given **rhsMatrix**. The size of the two matrices must be the same
subtract (rhsMatrix):	The same as the **add()** operation but subtracts the two matrices
multiply (rhsMatrix):	Creates and returns a new matrix that is the result of multiplying this matrix to the given **rhsMatrix**. The two matrices must be of appropriate sizes as defined for matrix multiplication

Scaling in Matrix

$$3\begin{bmatrix} 6 & 7 \\ 8 & 9 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3*6 & 3*7 \\ 3*8 & 3*9 \\ 3*1 & 3*0 \end{bmatrix} = \begin{bmatrix} 18 & 21 \\ 24 & 27 \\ 3 & 0 \end{bmatrix}$$

```
# Scales the matrix by the given scalar.
def scaleBy( self, scalar ):
  for r in range( self.numRows() ) :
    for c in range( self.numCols() ) :
      self[ r, c ] *= scalar
```

Matrix Addition

$$\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 6 & 7 \\ 8 & 9 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+6 & 1+7 \\ 2+8 & 3+9 \\ 4+1 & 5+0 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \\ 5 & 5 \end{bmatrix}$$

```
def __add__( self, rhsMatrix ):
  assert rhsMatrix.numRows() == self.numRows() and \
         rhsMatrix.numCols() == self.numCols(), \
    "Matrix sizes not compatible for the add operation."
   # Create the new matrix.
  newMatrix = Matrix( self.numRows(), self.numCols() )
   # Add the corresponding elements in the two matrices.
  for r in range( self.numRows() ) :
    for c in range( self.numCols() ) :
      newMatrix[ r, c ] = self[ r, c ] + rhsMatrix[ r, c ]
  return newMatrix
```

Matrix Subtraction

$$\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 6 & 7 \\ 8 & 9 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 - 6 & 1 - 7 \\ 2 - 8 & 3 - 9 \\ 4 - 1 & 5 - 0 \end{bmatrix} = \begin{bmatrix} -6 & -6 \\ -6 & -6 \\ 3 & 5 \end{bmatrix}$$

Matrix transposition

$$\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}^T = \begin{bmatrix} 0 & 2 & 4 \\ 1 & 3 & 5 \end{bmatrix}$$

Matrix Multiplication

$$A = \begin{bmatrix} A_{0,0} & A_{0,1} \\ A_{1,0} & A_{1,1} \\ A_{2,0} & A_{2,1} \end{bmatrix} \qquad B = \begin{bmatrix} B_{0,0} & B_{0,1} & B_{0,2} \\ B_{1,0} & B_{1,1} & B_{1,2} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix} * \begin{bmatrix} 6 & 7 & 8 \\ 9 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (0*6+1*9) & (0*7+1*1) & (0*8+1*0) \\ (2*6+3*9) & (2*7+3*1) & (2*8+3*0) \\ (4*6+5*9) & (4*7+5*1) & (4*8+5*0) \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 1 & 0 \\ 39 & 17 & 16 \\ 69 & 33 & 32 \end{bmatrix}$$

$$\begin{aligned}
C_{0,0} &= A_{0,0} * B_{0,0} + A_{0,1} * B_{1,0} \\
C_{0,1} &= A_{0,0} * B_{0,1} + A_{0,1} * B_{1,1} \\
C_{0,2} &= A_{0,0} * B_{0,2} + A_{0,1} * B_{1,2} \\
C_{1,0} &= A_{1,0} * B_{0,0} + A_{1,1} * B_{1,0} \\
C_{1,1} &= A_{1,0} * B_{0,1} + A_{1,1} * B_{1,1} \\
C_{1,2} &= A_{1,0} * B_{0,2} + A_{1,1} * B_{1,2} \\
C_{2,0} &= A_{2,0} * B_{0,0} + A_{2,1} * B_{1,0} \\
C_{2,1} &= A_{2,0} * B_{0,1} + A_{2,1} * B_{1,1} \\
C_{2,2} &= A_{2,0} * B_{0,2} + A_{2,1} * B_{1,2}
\end{aligned}$$

What should you know

- Differentiate between Array and Dynamic array
- Choose appropriately to use Array or Python’s List
- How to implement Array class
- How to implement 2D-Array class
- How to implement Matrix class